A Novel Stereo Vision Sensor for Fast Moving Objects

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ABSTRACT
In this paper, we propose a novel stereo vision sensor for the automatic 3-D dynamic surface measurement, which is able to simultaneously acquire the 2-D scan image and the 3-D reconstruction results. The proposed system mainly concludes a line scan camera, a frame camera and two line lasers. Due to the application of line lasers, the proposed system is able to work accurate and stable under complex lighting condition. About our system, the view plane of line scan camera and the light plane of line laser are in coincidence. First, the calibration of intrinsic, extrinsic parameters and the light plane coefficients are obtained. Epipolar line is calculated to extract the accurate position of image point corresponding to each pixel of line scan camera. Finally, the 3-D shape is reconstructed according to binocular stereo vision model. The 2-D image and 3-D fa ̧ade information can be synchronously acquired by scanning over objects. Physical experiments show that the proposed stereo vision sensor can provide robust and accurate results.

CCS Concepts
•Computer systems and organization→ Real-time systems → Real-time system architecture

Keywords
Dynamic measurement; stereo vision sensor; 2-D scan image; 3-D surface reconstruction.

1. INTRODUCTION
Inspection methods which based on the images and 3-D shape are widely utilized in various domains, including automobile [1], robotic [2], railway [3] and industry [4]. Up to now, tremendous professional and consumer-grade stereo vision sensors have been launched [5-6]. For the inspection methods which based on the 2-D images, the accuracy is seriously affected by the inevitable factors, such as complex lighting conditions, color, mud, grease. Obviously, the recognition accuracy would increase when the 3-D shape data are considered synchronously.

Existing 3-D shape measurement systems mainly adopt the frame camera. The image quality of frame camera is greatly affected under nonuniform lighting condition. Line scan camera is 1-D imaging sensors, which is simple in structure. With the advantages of high resolution and high line rate, line scan camera is quite suitable for visual inspection of fast moving objects, including automobile, train, workpieces on the production line, and so forth. With line lasers serving as auxiliary light source, line scan camera is able to obtain clear images of fast moving objects with low exposure time.

Based on the above analysis, a novel stereo vision sensor is presented in this paper. First, the intrinsic parameters of line scan camera and frame camera, extrinsic parameters between two cameras and light plane coefficients are calibrated. Nonlinear optimization method is utilized to improve the calibration accuracy. In this paper, we assume that the view plane of line scan camera and the light plane of line laser are in coincidence. Hence, the matched point in frame camera is solved as the intersection of epipolar line and light stripe center. Thereafter, the 3-D surface is reconstructed based on line-structured light vision sensor. In general, the proposed measurement system is rapid, accurate and stable in application.

The rest of this paper is organized as follows. In section 2, the related works of stereo vision sensor, line scan camera system and line scan camera calibration methods are introduced. In section 3, the mathematical model, calibration method and measurement principle of the proposed stereo vision sensor are presented in detail. The effectiveness of proposed system is verified physical experiments in Section 4. Conclusions are drawn in Section 5.

2. RELATED WORK
With the development of computer vision technologies, a huge number of 3-D visual measurement methods have been proposed. In general, existing 3-D shape measurement methods can be divided to two types: passive and active. The main disadvantage

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of passive vision measurement technology is that lighting condition, the temperature and texture of the object greatly influenced the accuracy. What's more, the passive vision measurement technology cannot obtain adequate information without auxiliary measurement methods.

Active vision measurement technology obtains 3-D surface shape according to the modulating action of the projected beam. Structured light measurement method is the most widely used and most mature 3-D vision measurement method. It is simple in equipment, automatic and flexible in choosing the light projector power and wavelength. As a result, it is very suitable for dynamic 3-D measurement under complex lighting condition.

Line scan camera calibration is an essential work in the application. Horaud [7] introduced the pioneer line scan camera calibration which based on several feature lines which locates on a planar pattern. The accuracy is limited for the lack of projective information. Consequently, calibration method based on 2-D scan images which are gathered through composing sequential linear image to achieve high calibration accuracy. For these methods, moving platforms are essential [8].

With further research in the model of line scan camera, studies of 3-D shape measurement using line scan cameras have been reported. A stereo line-scan syste for 3-D surface reconstruction is proposed in [9].The structured light methods time correlation and phase shifting are adopted to implement 3-D surface reconstruction. However, the depth range is very limited. Ilchev et al [10] proposed a high-speed line scan sensor based on a one-dimensional correction method. However, surface texture greatly affects its measurement accuracy. Denkena et al [11] proposed a single line-scan camera for measuring sheet metal parts. However, the result is greatly affected by the phase error caused that comes from the phase extraction algorithm and the surface contamination.

3. SYSTEM DESCRIPTION

3.1 System Configuration

The proposed measurement system mainly consists of line scan camera, frame camera, image acquisition system, control units, as shown in Figure 1.

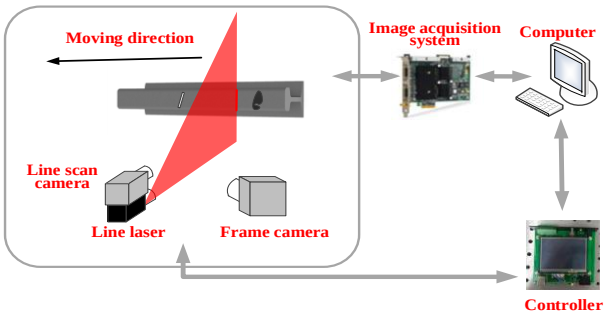


Figure 1. Structural schematic of the 3-D surface vision measuring system.

In essence, the proposed measurement system is a line-structured light vision sensor, and its mathematical model is established in Figure 2. Some notations are defined as follows.

1) K_l : Intrinsic parameters of line scan camera;
2) K_f : Intrinsic parameters of plane camera;
3) R_{lf}, t_{lf} : Structure parameters from line scan camera to frame camera;
4) R_l, t_l : Extrinsic parameters of line scan camera;
5) R_f, t_f : Extrinsic parameters of frame camera;
6) k_{l1} : First order distortion coefficient of line scan camera;
7) k_{f1}, k_{f2} : First and second order distortion coefficient of line scan camera;
8) $[a,b,c,d]$: Coefficients of light plane;
9) $O_{c1}x_{c1}y_{c1}z_{c1}$: The line scan camera coordinate system;
10) $O_{c2}x_{c2}y_{c2}z_{c2}$: The frame camera coordinate system.

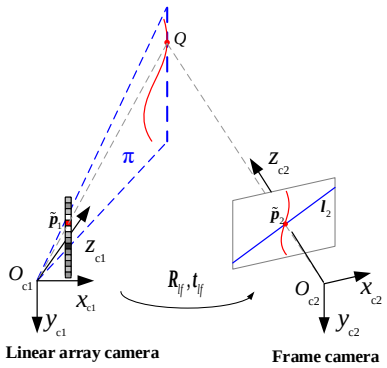


Figure 2. Line-structured light vision sensor.

The reconstruction equation of line-structured light vision sensor is established as follows,

$$\begin{cases} ax_f + by_f + cz_f + d = 0 \\ \rho \tilde{\boldsymbol{p}}_r = \boldsymbol{K}_f \left[\boldsymbol{I}_{3\times3} \quad \boldsymbol{0}_{3\times1} \right] \boldsymbol{Q}_f \end{cases} \tag{1}$$

where ρ is nonscale factor, $\boldsymbol{Q}_f = [x_f, y_f, z_f, 1]^{\mathrm{T}}$ are the 3-D homogeneous coordinates of $\boldsymbol{Q}$ under $O_{c2}x_{c2}y_{c2}z_{c2}$.

3.2 System Calibration

The calibration results of the measurement system have great influence on the measurement accuracy. The calibration of the proposed measurement system mainly includes three main parts: the intrinsic parameters calibration of two cameras, extrinsic parameters calibration of measurement system and the light plane coefficients calibration. First, the nonlinear optimal solution of K_f and the linear solution of $K_l, R_{lf}, t_{lf}, [a,b,c,d]$ are calculated. Then, nonlinear optimization method is utilized to obtain the optimal solution of $K_l, R_{lf}, t_{lf}, [a,b,c,d], k_{l1}$ .The camera intrinsic parameter is invariant. In the calibration of frame camera, the image noise and extracting accuracy of feature points have great influence on the calibration accuracy. Liu's method [12] which based on image point location uncertainty is adopted to calibrate the intrinsic parameters of the frame camera, and its calibration accuracy is at least three times that of Zhang's method [13]. The structure parameters calibration is essential in stereo measurement. In this paper, the linear solution of intrinsic parameters and structure parameters of linear camera is obtained by the Hui's method [14]. Thereafter, the light plane coefficients $[a,b,c,d]$ are obtained using the plane fit method.

Nonlinear optimization improve the accuracy of calibration results. Three objective functions are established. The first objective function is established with the minimum difference between the projected point and image points in the line scan camera, as shown in Equation (2):

$$\boldsymbol{e}_1(a_1) = \min(\sum_{i=1}^{M}\sum_{j=1}^{N} Dist(\tilde{\boldsymbol{p}}_{ij}, \boldsymbol{K}_l[R_l \ t_l]\boldsymbol{Q}_{Tj})) \quad (2)$$

where $a_1 = [\boldsymbol{K}_l, k_{l1}, \boldsymbol{R}_{lf}, \boldsymbol{t}_{lf}]$, M is the number of placements, N is the total number of feature points, $\tilde{\boldsymbol{p}}_{ij}$ is undistorted image points in line scan camera.

The objective function for optimization is established with the minimum difference between the reconstructed point of the feature points on the target and the actual location, as shown in Equation 3:

$$\boldsymbol{e}_2(a_2) = \min(\sum_{j=1}^{N} Dist(\boldsymbol{Q}_j, \boldsymbol{Q}_{Tj})) \quad (3)$$

where $a_2 = [a,b,c,d]$, $\boldsymbol{Q}_j$ is reconstructed coordinates of the feature point, $\boldsymbol{Q}_{Tj}$ is the actual location of the feature point.

The final optimization objective function is established as follows,

$$\boldsymbol{E}(a) = \boldsymbol{e}_1(a_1) + \boldsymbol{e}_2(a_2) \quad (4)$$

Thereafter, the nonlinear optimum solutions of calibration results are obtained by the LM method.

3.3 Measurement Principle

In general, the measurement steps of proposed system are concluded as follows.

Step 1: Line scan camera captures the moving objects within a given clock rate, meanwhile, frame camera obtains the light stripe images.

Step 2: The epipolar line is calculated corresponding to each pixel in line scan camera. The match point is determined as the intersection of epipolar line and light stripe.

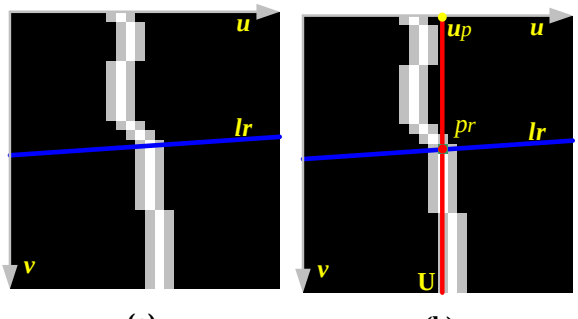
Step 3: The 3-D coordinates are calculated according to the binocular stereo vision sensor model. Thereafter, the coordinate along x axis is determined by the product of the acquisition interval and the time of acquisition.

Step 4: In push-broom mode, the proposed system is able to acquire 2-D scan image and 3-D façade information synchronously and continuously.

The extraction accuracy of stripe center points affects the measurement accuracy. Steger method [15] is the most widely utilized method to extract the center of light stripe for its high accuracy and generality in application. Hence, Steger method is adopted to extract the matched point in frame camera. According to the epipolar constraint, each image point $\boldsymbol{p}_l$ is corresponding to a epipolar line l_f . For $\boldsymbol{p}_l$, the matched image point $\boldsymbol{p}_r$ in frame camera is determined by the intersection of l_f and light stripe in frame camera. Hence, we calculate l_f by the following equation:

$$\boldsymbol{l}_f = \boldsymbol{F}\boldsymbol{p}_l \quad (5)$$

where $\boldsymbol{F}$ is the foundation matrix, $\boldsymbol{F} = \boldsymbol{K}_f^{-T}\boldsymbol{S}_{fl}\boldsymbol{R}_{fl}\boldsymbol{K}_l^{-1}$. $\boldsymbol{S}_{fl}$ is the antisymmetric matrix of t_{fl} .The extraction process of the matched point is shown in Figure 3.



(a) (b)

Figure 3. Extraction process of the matched point. (a) The location of epipolar line. (b) Extraction of the accurate subpixel coordinates of the match point in frame camera.

Figure 3(a) shows the location of epipolar line. On the light stripe, the subpixel coordinates of points $\{p\}$ which has the biggest grayscale are obtained by Steger method. Thereafter, the matched point $p_r = (u_p, v_p)$ is determined by the center point which is closest to the corresponding epipolar line. The extraction results based on our method is shown in Figure 4.

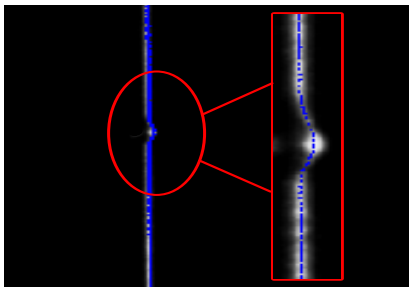


Figure 4. Extraction result of the match points on light stripe.

As shown in Figure 4, our method is able to extract points accurately even the shape and width of light stripe varies.

4. EXPERIMENTAL RESULTS

In physical experiments, the effectiveness of proposed system is validated by measurement error. As shown in Figure 5, two cameras are equipped with the same 17 mm optical lens. The image resolution of line scan camera is 1×2048 pixels, the image resolution of frame camera is 2336×1728 pixels, the field of view is approximately $600\text{mm}\times500\text{mm}$, and the distance of measurement is approximately 600 mm. Infrared laser(808nm) provides illumination source for line scan camera, visible laser (635nm) is utilized to locate the view plane of line scan camera. What's more, a mechanical regulation structure is adopted to regulate the positions of line lasers and line scan camera precisely.

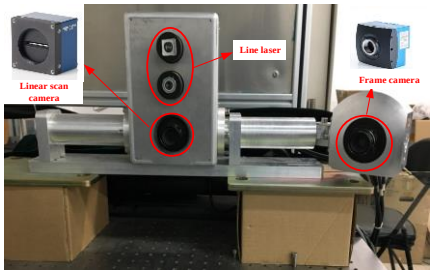


Figure 5. Stereo vision sensor in the physical experiment.

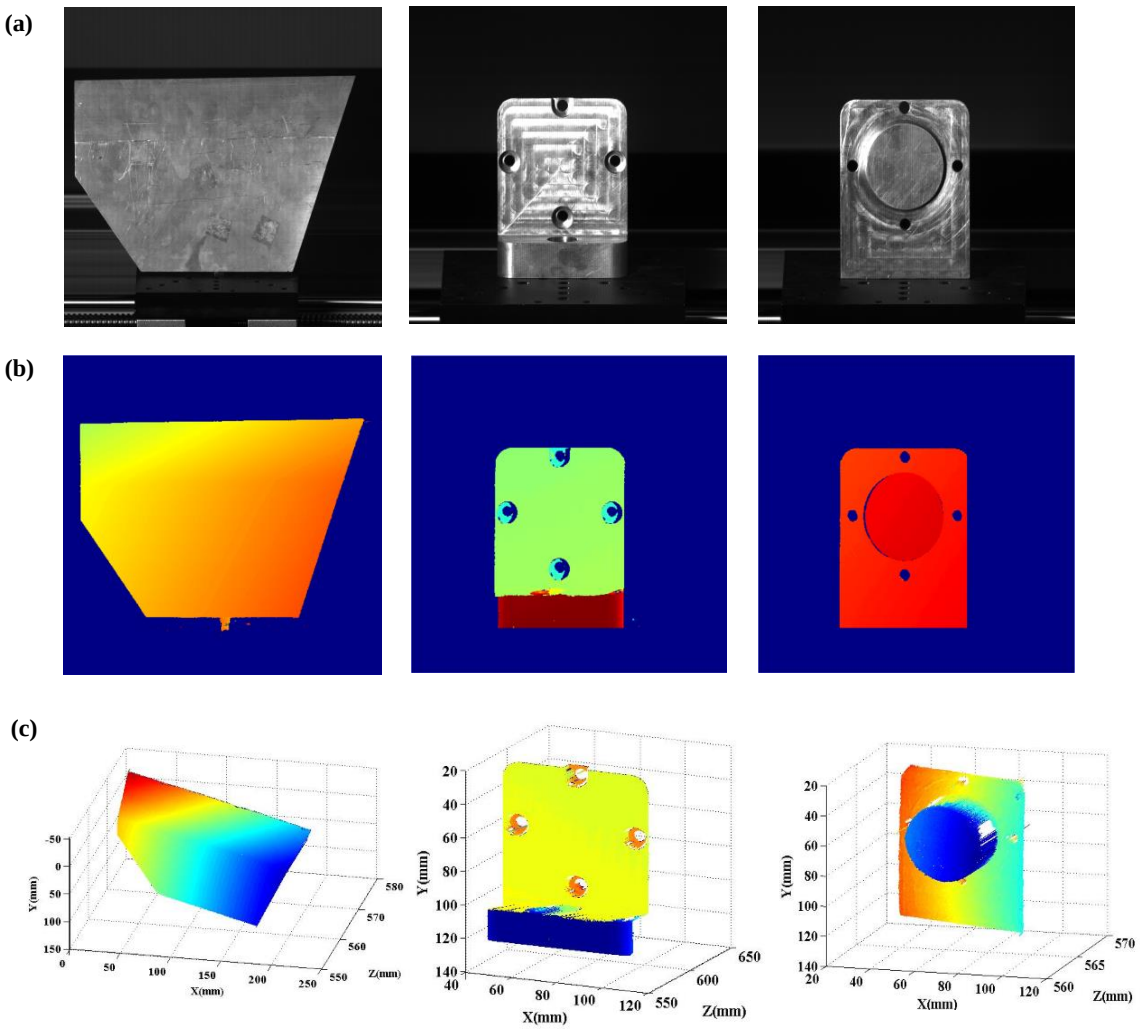


Figure 6. Measurement results using the proposed system. (a) The 2-D scan images obtained by line scan camera; (b) Depth images of workpieces; (c) The 3-D surface reconstruction results of workpieces.

The calibrated intrinsic parameters of two cameras are shown in Table 1.

Table 1. Intrinsic parameters calibration results.

	f_x(pixel)	f_y (pixel)	u_0 (pixel)	v_0 (pixel)
Frame camera	2532.21	2532.11	1166.09	631.99
Line scan camera	-	1030.35	-	632.18

The distortion coefficients of line scan camera and frame camera are $k_{f1}=-0.25, k_{f2}=0.48, k_{l1}=-0.2$, respectively. The calibration result of light plane is $1.5689x_f-0.0478y_f-567.0871z_f+1=0$. The extrinsic parameters are:

$$R_{lf}=\begin{bmatrix}0.843 & 0.015 & 0.538\\ 0.025 & 0.999 & 0.012\\ -0.537 & -0.024 & 0.843\end{bmatrix}, t_{lf}=\begin{bmatrix}-303.764\\ -13.594\\ 133.263\end{bmatrix}$$

To evaluate the measurement accuracy, we reconstruct the surface of standard workpiece. For the workpiece, the distance of the two parallel planes is 25mm . The measurement error Δd is denoted as the deviation between measurement distance d_m and ideal distance d_t .The methods are tested 100 times adopting each calibration results. The measurement errors are analyzed as follows. The RMSE of measurement distance RMSE_M is calculated by Equation 6:

$$\text{RMSE}_M=\sqrt{\frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}\Delta d_{ij}^{\ 2}} \tag{6}$$

where m is the times of measurement trial, and n is the total number of the measurement errors obtained in each measurement trial. Thereafter, the RMSE_M of proposed method is calculated as 0.28mm , which fulfill the actual demands in application. Under the same experimental conditions, we calculated the RMSE of a line-structured light vision sensor proposed in [16]. The RMSE_M of Liu's method is 0.35mm , that is, the proposed system is relatively accurate.

To further evaluate the effectiveness of the proposed system, we reconstructed the 3-D surface of several workpieces as shown in Figure 6. Figure 6(a) shows the 2-D scan image captured by line scan camera. The scan images are of high quality. After 3-D reconstruction, we take the *z* coordinate of each point as the depth corresponding to the pixel in line scan camera. Thereafter, depth images can be obtained as shown in Figure 6(b). Figure 6(c) shows the 3-D surface reconstruction results of the workpieces. For these workpieces, the surface reconstruction results are in accordance with the actual shapes. That is, our system is reliable in application.

5. CONCLUSION

This paper introduces a novel 3-D surface measurement system which is able to synchronously obtain the 2-D scan image and the 3-D surface of the fast moving objects. In detail, the proposed system is suitable for various applications including the defect detection of paper, the inspection of missing split pin in railway, the recognition and tracking for moving objects and so on. The proposed system is mainly composed of line scan camera, frame camera and line lasers. From the experiments, the measurement accuracy of the proposed method is relatively high. With a field range of about $600\text{mm} \times 500\text{mm} \times 300\text{mm}$, the proposed system can achieve a measurement accuracy of 0.28mm, which is appropriate in the visual inspection. In the fact, the view plane of line scan camera and the light plane of line laser are not in coincidence absolutely. In the further improvement, we will focus on the compensation of the error caused by that factor.

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